Artificial Intelligence

VICTORIA UNIVERSITY OF WELLINGTON
TE HERENGA WAKA

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Machine Learning Fundamentals

Dr Fangfang Zhang

SECS & Centre for Data Science and AI

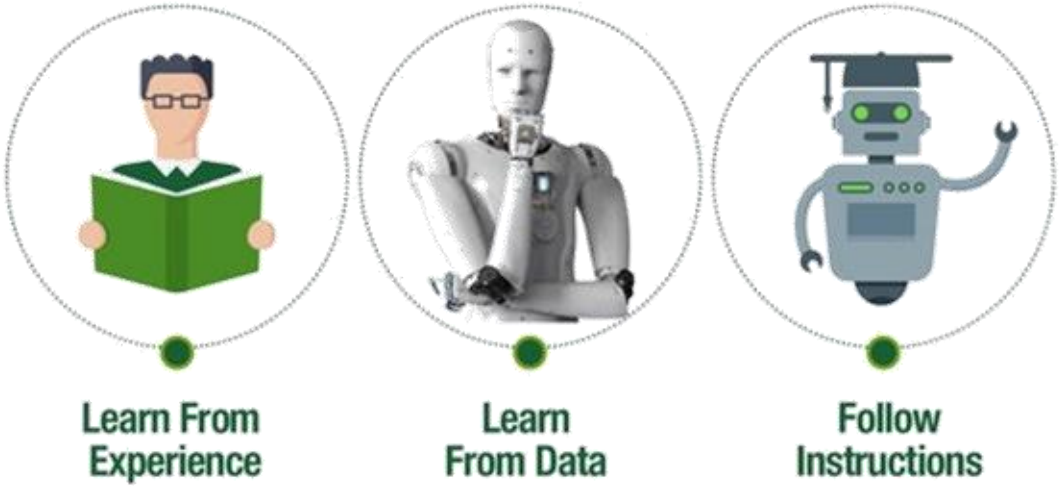
fangfang.zhang@vuw.ac.nz

Outline

- What is Machine Learning
- Why Machine Learning
- Types of Machine Learning
- Machine Learning Tasks
- Data: training set vs. test set
- Generalization

What is Machine Learning

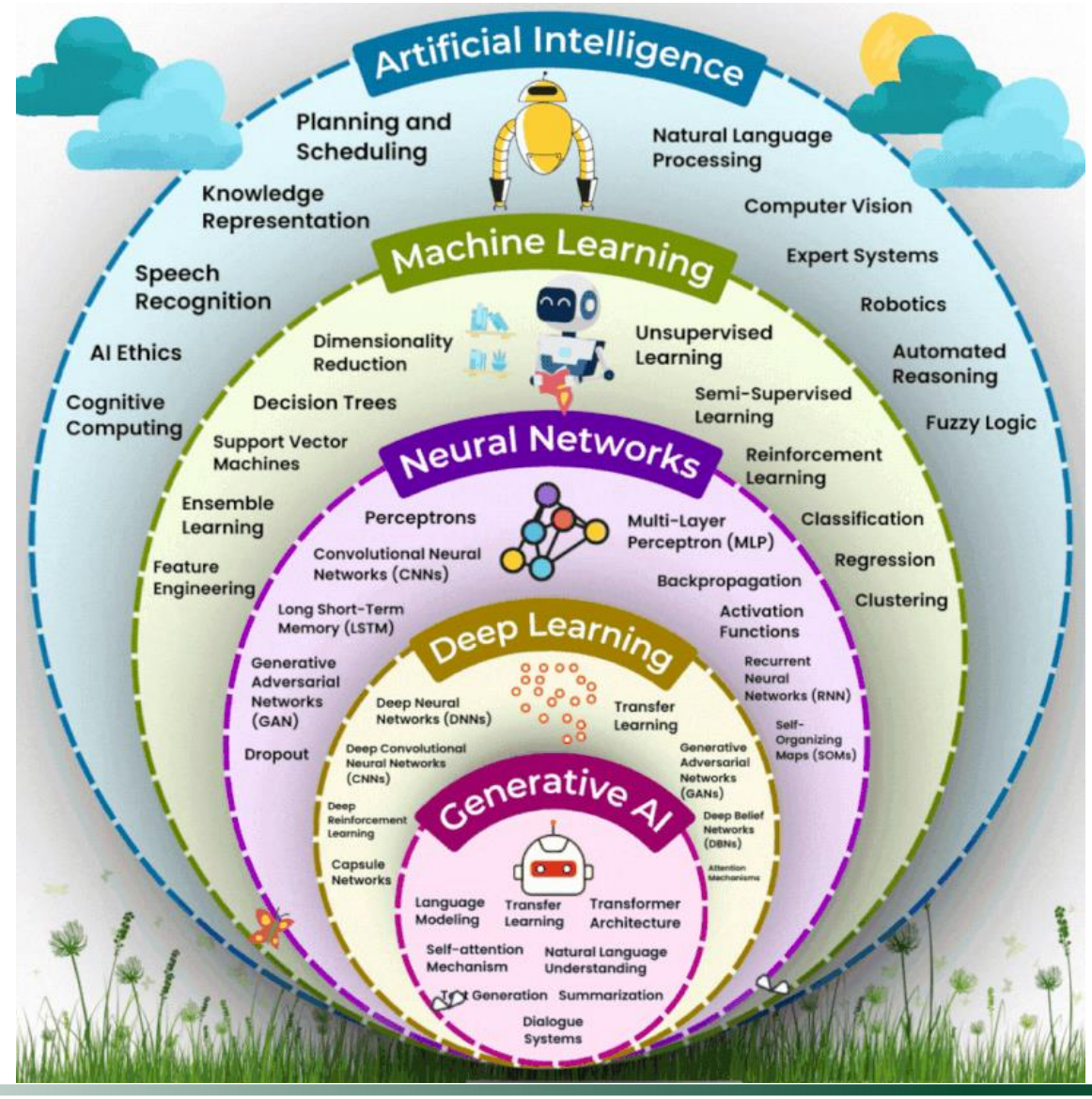
- Machine learning is concerned with the **design and development** of algorithms/techniques that allow computers to "learn"
- "Machine learning is the study of computer algorithms that improve automatically through **data** (experience)"
- Systems learn to **identify patterns** and make decisions with minimal human intervention



Why Machine Learning

- To make smarter machines (systems) for decision making
  - Automation, save efforts of human
  - Improve performance, without (or with little) human intervention
  - Efficiency, quick decision making
  - Robust behavior in noisy environments
- To process:
  - Image: Sora, Digit recognition, face recognition
  - Text: ChatGPT, Gemini, DeepSeek
  - Speech: Siri, compose music, …
  - Simulation: train scheduling system
  - Tabular data: https://archive.ics.uci.edu/datasets

The AI Universe



Types of Machine Learning

Supervised learning*

- E.g., Regression, Classification
- Predictive task, e.g., predict the house price, number of enrolments
- **The most widely explored** type of machine learning

Unsupervised learning*

- E.g., Clustering, and Association Rules
- Exploratory task, e.g., customer segmentation & marketing

Others

- Semi-supervised learning
- Reinforcement learning

** Disclaimer: There are several other tasks that qualify as "supervised" or "unsupervised" for now, for example, unsupervised classification. Let's stick to these for simplicity in the course context*

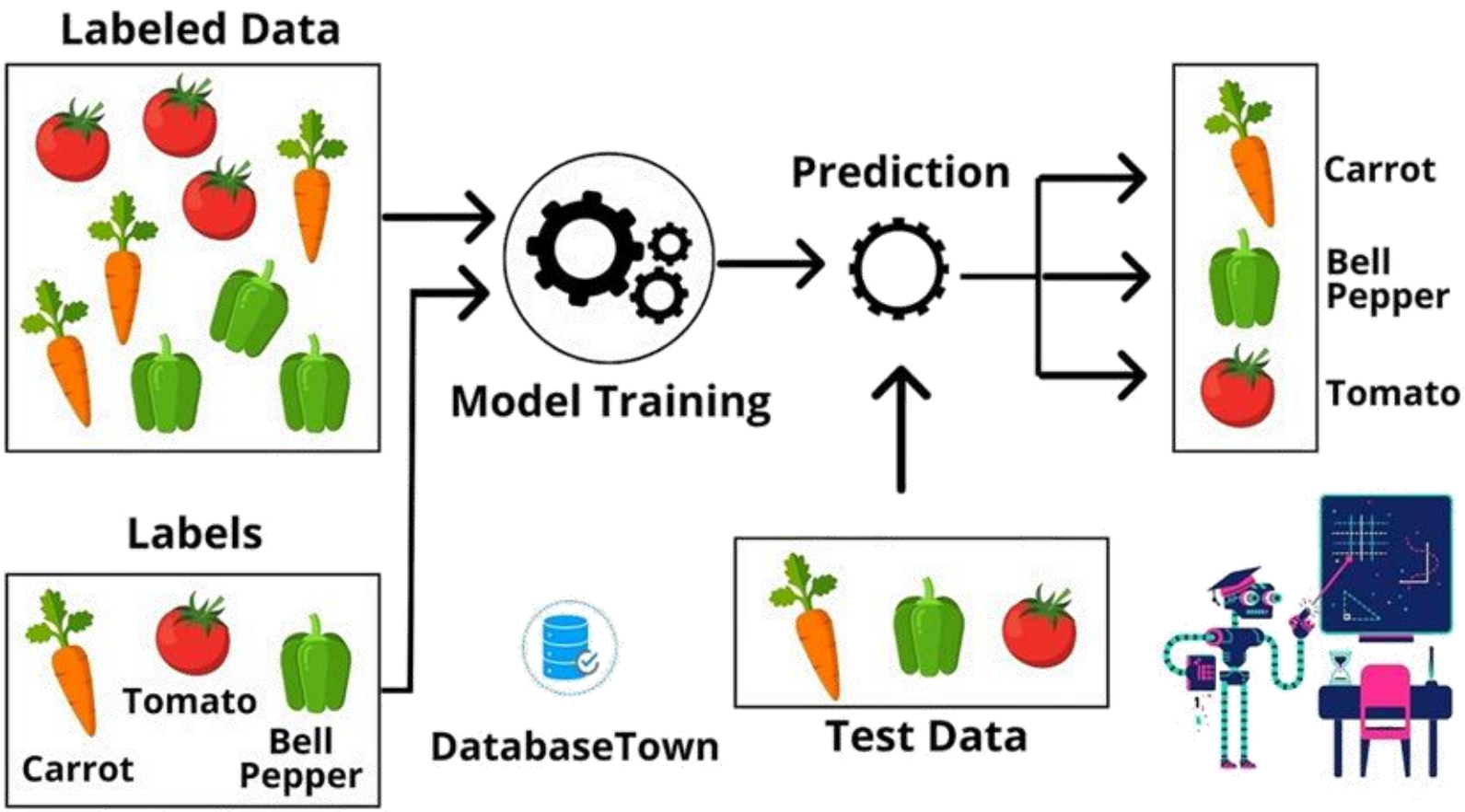
Supervised Learning

- **Given**: instances of inputs and target outputs (**labels**)
- **Generate**: a model that maps inputs to desired outputs
- **Predict**: the correct output for a **new (unseen)** input
- Examples:
  - Learn to recognise words from handwriting documents
  - Learn patterns/trends for predicting the stock market/weather/traffic
  - Learn patterns from fingerprint for terrorist detection at airports

House ID	Square Footage	Bedrooms	Neighborhood	Year Built	Sale Price (Target)
H-001	1,200	2	Suburb A	1995	$250,000
H-002	2,500	4	Downtown	2015	$650,000
H-003	1,800	3	Suburb B	2005	$410,000
H-004	950	1	Downtown	1980	$320,000
H-005	3,200	5	Suburb A	2022	$890,000

Supervised Learning

General process of machine learning:



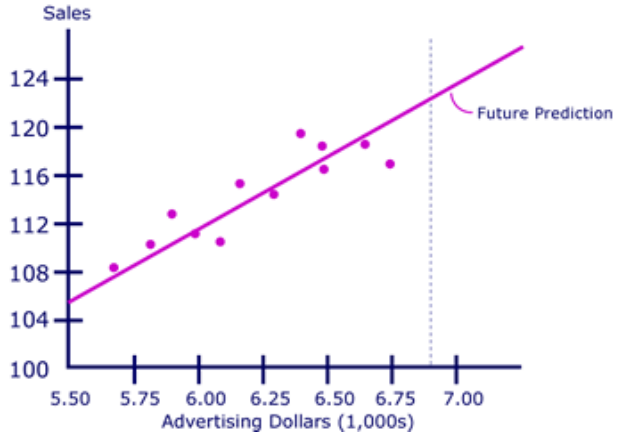
Other Learning Types

- **Unsupervised Learning**
  - Given: set of **unlabeled** instances
  - Infer: subsets of "similar" instances (disjoint or overlapping)
  - Examples:
    - Find clusters in high-dimensional data
    - Identify groups of genes that have similar properties
- **Semi-supervised learning**: A mixture of supervised learning and unsupervised learning
- **Reinforcement Learning**:
  - Given: Sequence of actions and states, and reward/penalty occasionally
  - Policy for choosing best actions
  - Examples: Robot navigation tasks, Multiple lift controller, ...

Machine Learning Tasks

Regression:

- Map a data to a real-valued prediction variable
- Supervised learning
- Learning a function, i.e., regression model
- Assume a certain function type (e.g. linear, logistic, polynomial, …) and determine the best function of this type to fit the given data
- Learn the function type at the same time (Symbolic Regression)

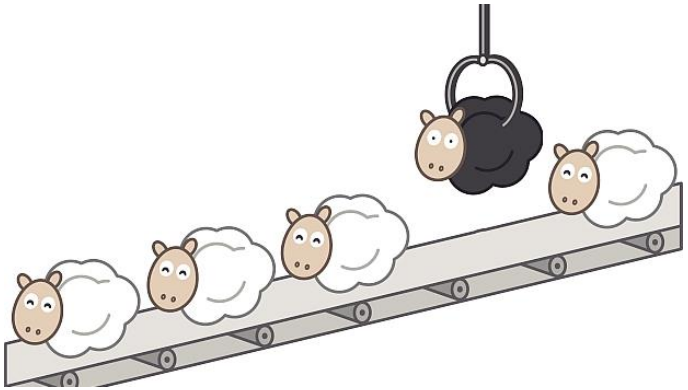


- Examples
  - Financial prediction
  - Saving prediction
  - Ad cost vs sales

Machine Learning Tasks

Classification:

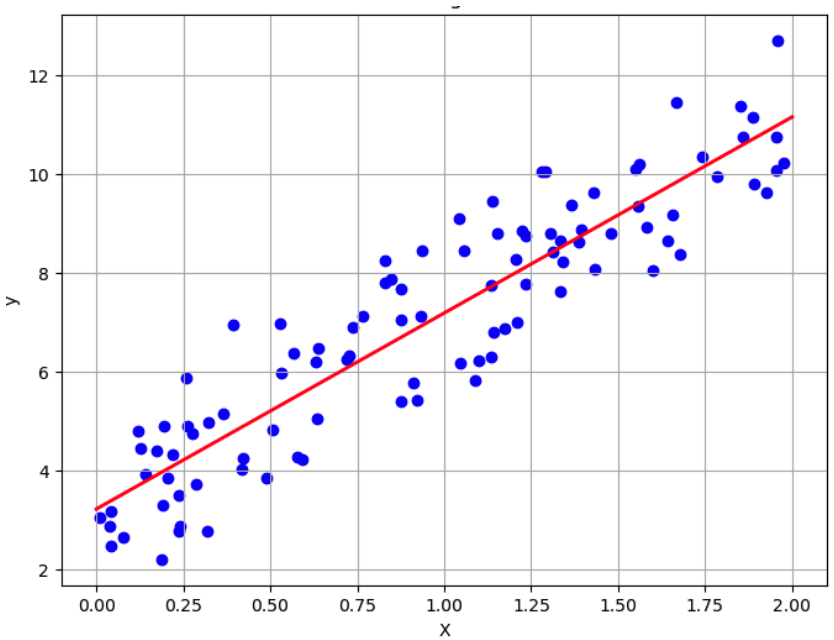
- Maps data into predefined groups or classes
- Supervised learning
- Learn a classifier to make decision



- Examples
  - Medical: cancer vs not cancer
  - Bank: credit reliable vs unreliable
  - Digit recognition: multi-class
  - Weather: sunny or rainy
  - Anomaly detection
  - Airport security screening: terrorist/criminals or not

Machine Learning Tasks

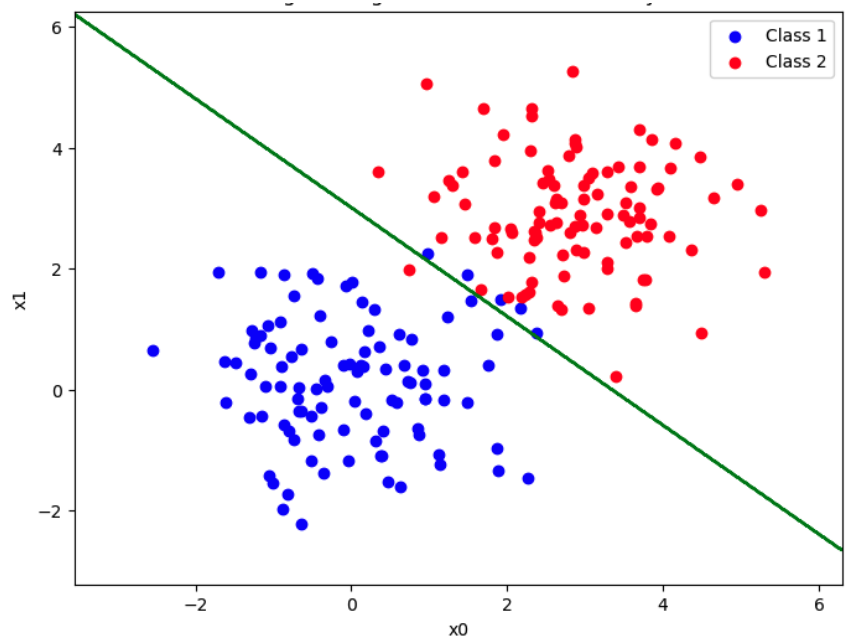
Regression



Map into real-valued prediction variable

- **Output:** a **continuous value**

Classification



Maps into predefined groups/classes

- **Output**: possible class label

A regression task and classification task? depends on the “targets”

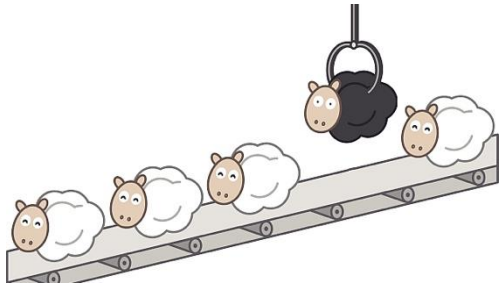
Machine Learning Tasks

Regression

- **Predicting the price of a used car** based on its age, mileage
- **Forecasting the sales of a product** next month based on historical sales data
- **Predicting energy consumption** based on weather conditions and occupancy

Classification

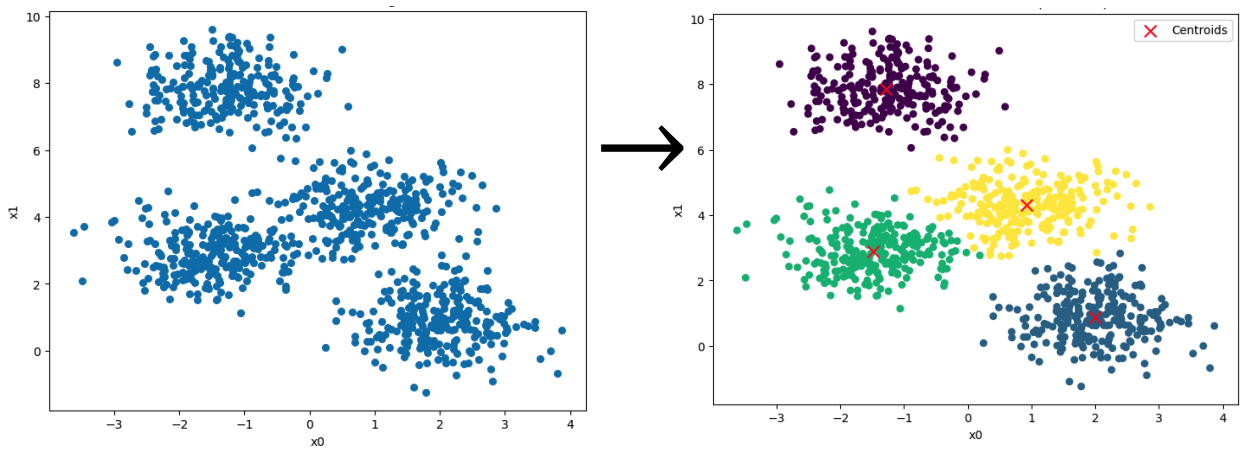
- **Identifying whether an email is spam or not** based on the content of the email and sender information
- **Classifying handwritten digits** (0 to 9) based on the pixel information in an image
- **Determining whether a customer is likely to churn** (cancel their subscription) based on their past purchase history and demographics



Machine Learning Tasks

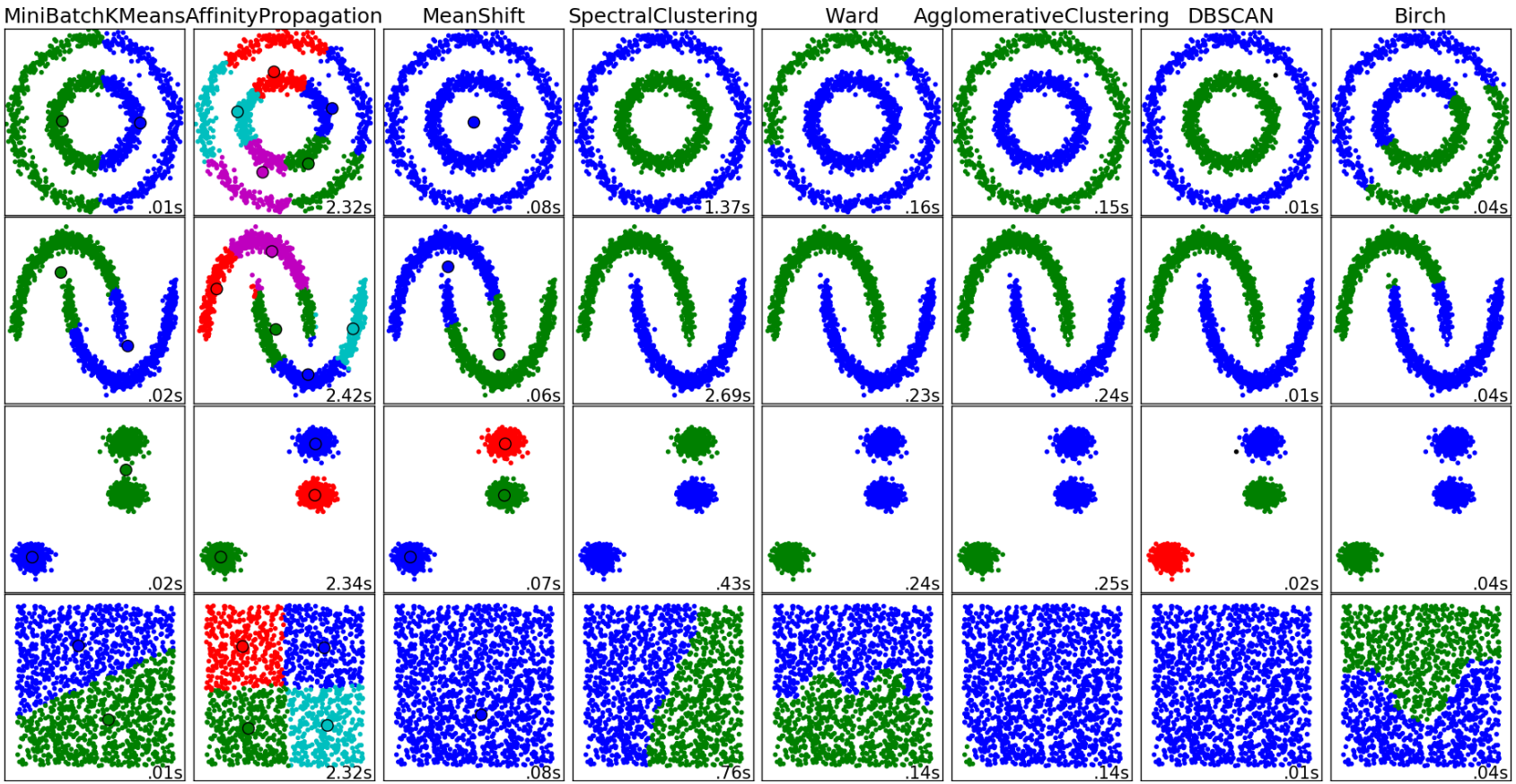
Clustering:

- Similar to classification, **except that the groups are not predefined (not labelled)**, but rather defined by data itself
- Unsupervised learning
- Models segment or partition data to form groups based on the similarity/distance
- A domain expert is needed to interpret the meaning



Machine Learning Tasks

Clustering:



Association Rules

- Classic data mining techniques
- Uncover relationships among data
- Often used in the retail sales community to identify items that are frequently purchased together
- Association occurrence
- Applications:
  - Optimize product placement, recommendation, …

"**if** *a customer buys bread*, they are **also likely** to buy *milk*"

or simply:

$$bread \rightarrow \boxed{milk}$$

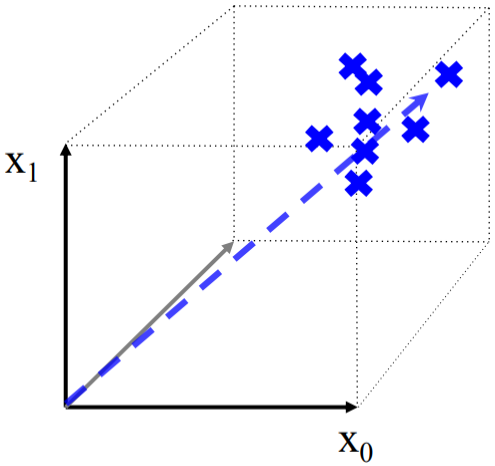
Association Rules

- Beer & Nappies

Data

- Foundational input of machine learning algorithms
- Presented with some positive and some negative for classification tasks

V1	V2	V3	V4	Class
3.6216	8.6661	-2.8073	-0.44699	1
4.5459	8.1674	-2.4586	-1.4621	1
3.866	-2.6383	1.9242	0.10645	1
3.4566	9.5228	-4.0112	-3.5944	1
0.32924	-4.4552	4.5718	-0.9888	1
4.3684	9.6718	-3.9606	-3.1625	1
3.5912	3.0129	0.72888	0.56421	1
2.0922	-6.81	8.4636	-0.60216	1
3.2032	5.7588	-0.75345	-0.61251	1
1.5356	9.1772	-2.2718	-0.73535	1
1.2247	8.7779	-2.2135	-0.80647	1

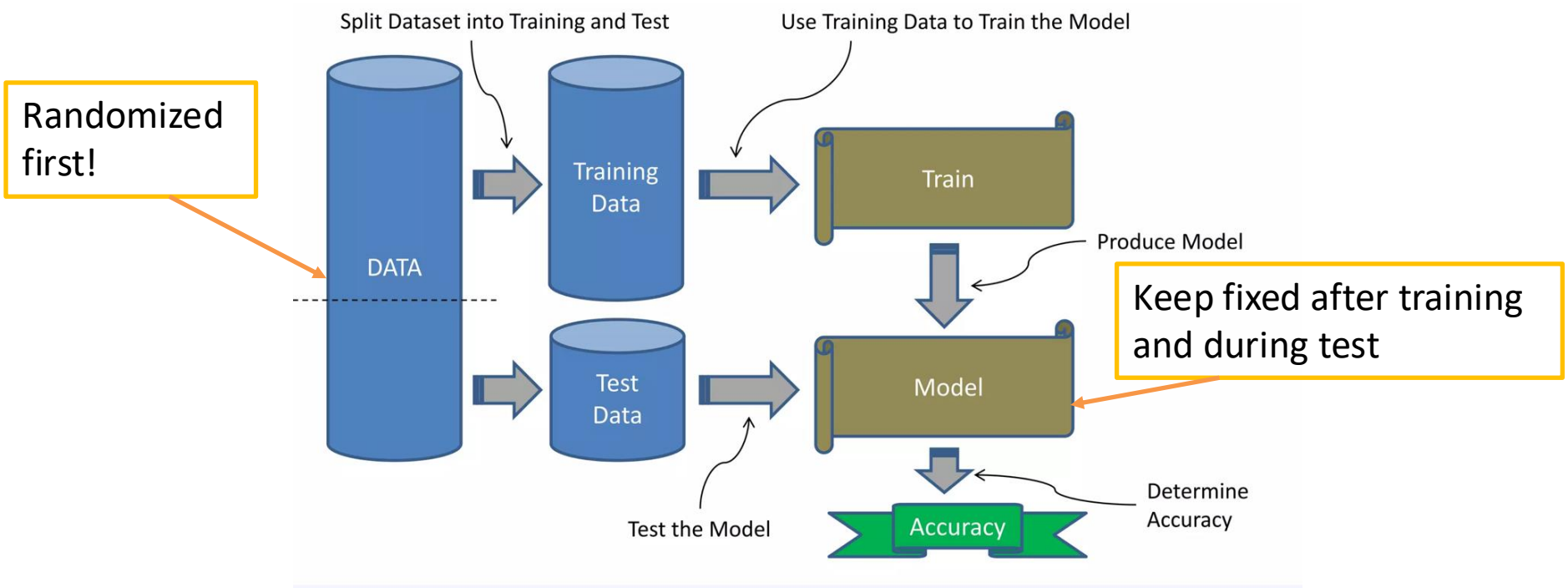


Data as Points

Humans can't see in more than 3D.
Some of our intuitions hold, some fail

Dataset Splitting

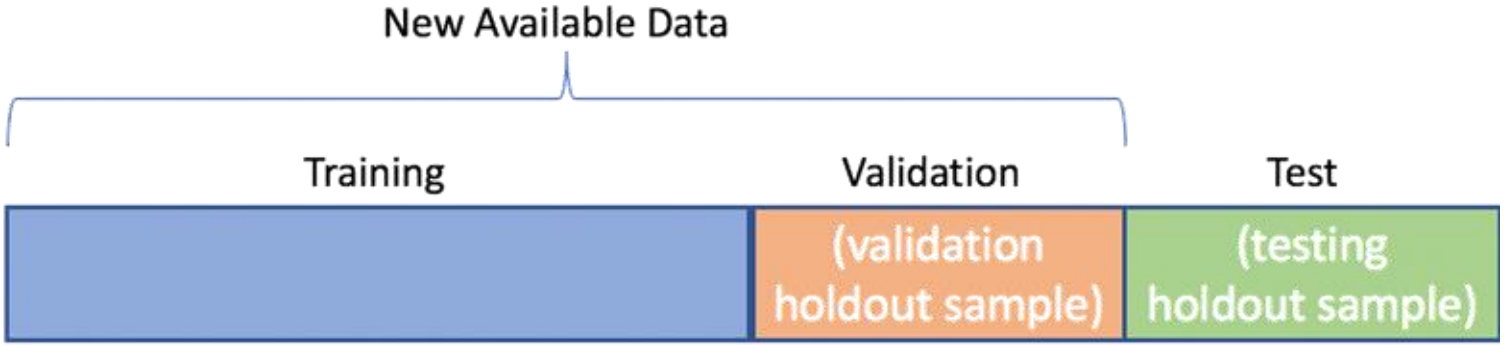
- **Training set**: a collection of instances from which a model is trained
- **Test Set**: A collection of instances which were never used for learning the model (otherwise, cause data leakage issue)
  - For measuring the performance of the learnt classifier



Dataset Splitting

- Training set VS Test set

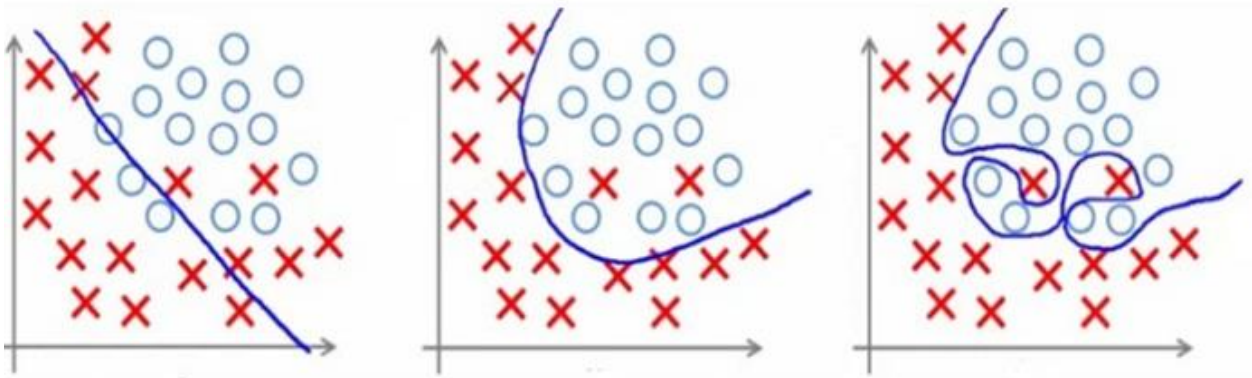
- Training set, **Validation set**, and Test set
  - Not used for training model and not used for test
  - Unseen data for learned model
  - But can be used during the training process, e.g., model selection, improve the process of training model



Generalization

- We learn a classifier/predictor/model from the training data
- But performing well on training data is **NOT** enough!
- Important to evaluate the performance on the **test (unseen) data** --- **generalization**

- If too biased to the training data, this may cause *overfitting*: too good on the training data, but poor on test data



Summary

- Basic concepts of machine learning
  - Supervised learning and unsupervised learning
- Common machine learning tasks
  - Regression
  - Classification
  - Clustering
- Data
  - Training data VS test data
  - Dataset Splitting
- Generalization

Next:

- Tutorial 1